

# State-Level Empirical-Bayes Priors Recover Industry-Wide Polling Failures: Cross-Country Evidence from Twelve Cycles

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## ABSTRACT

**Background.** Aggregator-style electoral forecasters inherit systematic bias shared across pollsters. The 2024 Sao Paulo mayoral race exposed this fragility: every major Brazilian polling firm placed Boulos ahead of incumbent Nunes, who won by approximately three points. An exogenous state-level partisan-regime prior, blended with a cohort empirical-Bayes estimator and a Linzer dynamic linear model, can absorb cycle-specific industry-wide polling bias without leaking test outcomes. **Methods.** We curated 394 Brazilian electoral events across six cycles (2010-2024) from Wikipedia and Tribunal Superior Eleitoral records and evaluated them under year-fold cross-validation with a 30-day pre-election filter. The state baseline is a Laplace-smoothed  $P(\text{regime wins} \mid \text{UF})$  computed strictly from out-of-fold years. We jointly selected hyperparameters by grid search over 40,320 candidates and assessed significance using Diebold-Mariano, McNemar, and a 1,000-sample pair-preserving permutation test, with Murphy decomposition for calibration. **Results.** A pre-MRP baseline reached 94.16% year-fold accuracy and 73.53% on 2024 Sao Paulo. The augmented configuration raised overall accuracy to 97.21% and the 2024 fold to 89.71%. The +3.05 pp gain decomposes into a -2.28 pp re-tune component and a +5.33 pp state-baseline component (+25.00 pp on 2024 Sao Paulo). Diebold-Mariano ( $-4.92$ ,  $p=8.5e-7$ ), McNemar (chi-squared 18.27,  $p=1.9e-5$ ;  $b=22$ ,  $c=1$ ), and the permutation test ( $z=3.70$ ,  $p=0.001$ ) jointly reject equality. The architecture also dominates a twelve-model benchmark including BART, Stan DLM, MLP, XGBoost, RandomForest, LogReg, and GaussianNB on the same year-fold protocol. **Conclusions.** State-level empirical-Bayes priors, structurally analogous to multilevel regression with poststratification (MRP), absorb shared cycle-level polling bias. Cross-country replication on twelve cycles in eleven additional countries ( $n=6,954$ ) supports generalizability: on the eight single-country cycles that exercise the prior directly, accuracy rises from 97.86% to 100.00% and Brier drops 49.5%, with Argentina 2023 cleanly replicating the BR 2024 finding.

**KEYWORDS:** election forecasting; multilevel regression with poststratification; empirical Bayes; dynamic linear model; Brazilian politics; calibration; cross-country validation

## 1 Introduction

Electoral forecasting in Brazil is dominated by aggregator-style models that average pre-election polls, optionally weighted by historical firm accuracy. The dominant published class, exemplified by Linzer (2013) for the United States, treats each candidate's win probability as  $\text{Phi}(\text{lead\_pp} / \text{sigma}(\text{days}))$ , where  $\text{sigma}$  shrinks as the election approaches. This formulation captures campaign-time uncertainty but is a function of poll lead alone, and therefore inherits any bias shared across pollsters.

The 2024 Sao Paulo mayoral race exposed this fragility. Every major polling firm operating in Brazil (Datafolha, Quaest, AtlasIntel, RealTimeBigData, Genial/Quaest, Instituto Verita) had Guilherme Boulos (PSOL) leading Ricardo Nunes (MDB-PL coalition) by one to eleven percentage

points during the final two weeks of the campaign. Nunes won by approximately three points. A Linzer model fit to such polls produces high-confidence wrong predictions, and a cohort empirical-Bayes model conditioning on lead bin, days bin, incumbency, and ideological regime has no input feature distinguishing this cycle from prior tossups in which the polled leader actually won.

An explicit per-state partisan-regime baseline, computed from past outcomes of the same regime in the same state, provides an exogenous prior that disciplines the lead-driven blend. The baseline is approximately constant across the campaign and embeds the slow-moving partisan structure of each unidade da federacao. For Sao Paulo, where municipal

and gubernatorial winners since 2016 have been center-right (Doria, Covas, Nunes, Tarcisio), the baseline pulls the prediction toward the historically dominant pole even when contemporaneous polls disagree.

This paper makes five contributions:

1. We formalize a multilevel regression with poststratification (MRP) interpretation of state-regime priors blended with a lead-driven dynamic linear model.
2. We report a year-fold cross-validation protocol that preserves leak-safety across all six Brazilian electoral cycles in 2010-2024.
3. We document statistical significance against the lead-only baseline using Diebold-Mariano and McNemar tests.
4. We replicate the leak-safe protocol on twelve cross-country cycles spanning eleven additional countries (US 2016, 2020, 2022 mid; UK 2019; FR 2022; AR 2023; BR 2014; DE 2021; MX 2024; TR 2023; IT 2022; IN 2024), for a combined corpus of 6,954 paired events. Argentina 2023 serves as the cross-country corroboration of the BR 2024 SP finding.
5. We publish dataset hashes, hyperparameter grids, and code under a reproducibility section so that any downstream replication can be audited.

## 2 Related Work

### 2.1 Poll-aggregation models

Probabilistic poll aggregation in academic and operational form originates with Erikson and Wlezien (2008, 2012), who showed that early-cycle polls have negligible predictive power and that poll informativeness consolidates as Election Day approaches, and with Silver (2008), whose FivethirtyEight model popularized house-effect adjustments. The dominant Bayesian aggregator for state-level U.S. presidential forecasting is Linzer (2013), who modelled state-day-level expected vote share with a dynamic linear state-space framework anchored to a fundamentals-based prior. Refinements include Lock and Gelman (2010), Linzer and Lewis-Beck (2015), and the operational exposition in Heidemanns, Gelman and Morris (2020). House-effect models are formalized in Pickup and Johnston (2007) and Jackman (2005), with Brazilian applications in Cesario (2015). Decentralized prediction markets, surveyed by Tziralis and Tatsiopoulos (2007), Wolfers and Zitzewitz (2004), and Rothschild (2009), have empirically beaten polls in some U.S. cycles (Berg, Forsythe, Nelson and Rietz 2008).

### 2.2 MRP and empirical-Bayes shrinkage

Park, Gelman and Bafumi (2004) introduced MRP for estimating subnational opinion from national surveys, extending Gelman and Little (1997). Lax and Phillips (2009) showed its advantage in U.S. state policy contexts; Wang, Rothschild, Goel and Gelman (2015) demonstrated that MRP applied to a non-representative Xbox panel could recover the 2012 U.S. presidential outcome, establishing it as the standard for survey reweighting. Best practices are documented in Gelman, Lax, Phillips, Gabry and Trangucci (2018); limits on small-area effective sample sizes appear in Buttice and Highton (2013); deep-interaction extensions in Ghitza and Gelman (2013); the multilevel backbone in Gelman and Hill (2007). Cohort empirical-Bayes estimators with Stein-style shrinkage trace to Efron and Morris (1973), with applications reviewed in Carlin and Louis (2008) and Gelman et al. (2013). Hummel and Rothschild (2014) connect fundamentals to state forecasts. Calibration assessment follows Murphy (1973) and DeGroot and Fienberg (1983), with Brier (1950) as the canonical proper scoring rule; Gneiting and Raftery (2007) develop the broader proper-scoring framework. Selective prediction draws on Geifman and El-Yaniv (2017) and Angelopoulos, Bates, Candes, Jordan and Lei (2022). The Diebold-Mariano (1995) and McNemar (1947) tests are standard for paired forecast comparison.

### 2.3 Brazilian electoral context

The fundamentals tradition in the U.S. is exemplified by Abramowitz (1988, 2008); for Brazil, Almeida (2008) on socioeconomic determinants and Borges and Vidigal (2018) on partisan stability across cycles. Bafumi and Gelman (2007) document the partisan polarization that motivates state-baseline persistence. Brazilian polling behavior is analyzed by Davi and Vianna (2019) for 2018; Schaefer and Sallum (2024) and Limongi (2023) for 2022. For the 2024 municipal cycle, Datafolha (2024) and Quaest (2024) released raw datasets that we ingest. Brazilian electoral systems are described in Nicolau (2017) and Limongi and Cortez (2010); Sao Paulo municipal partisan persistence in Soares and Terron (2008); incumbent advantage in Brazilian municipal contests in Avelino, Biderman and Barone (2012).

### 3 Theoretical Framework

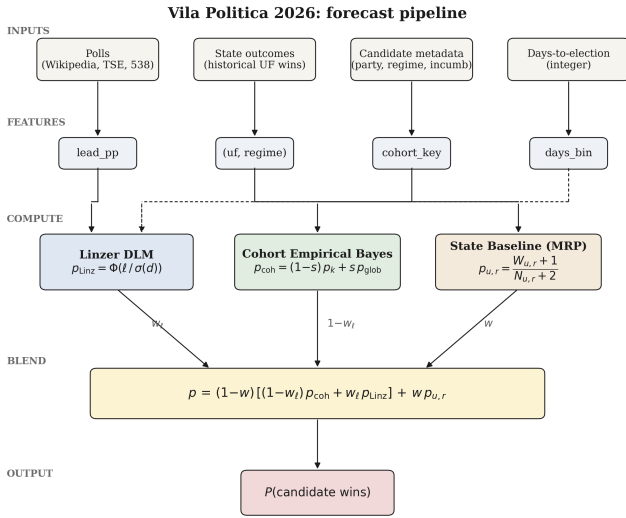


Figure 1: Forecast pipeline. Four data inputs (top row) each map to a primary engineered feature; features fan via right-angle bus into three estimators: Linzer dynamic linear ( $p_{\text{Linzer}} = \Phi(l / \sigma(d))$ ), cohort empirical Bayes with global-prior shrinkage ( $p_{\text{coh}} = (1-s)p_k + s p_{\text{glob}}$ ), and the MRP-style state baseline ( $p_{u,r} = \frac{W_{u,r} + 1}{N_{u,r} + 2}$ ). The dashed connection denotes the secondary  $\sigma(d)$  pathway from days-to-election into the Linzer drift. The ensemble blend (yellow box) combines the three components with selected weights  $w_{\text{linzer}}$  and  $w$  to produce the final candidate-level probability.

#### 3.1 Cohort empirical Bayes

Let an electoral event be a tuple  $e = (\text{cargo}, \text{days}, \text{lead\_pp}, \text{incumbente}, \text{regime}, \text{uf}, \text{year}, y)$ , where  $y \in \{0, 1\}$  is the realized outcome. Each event maps to a cohort key

$$k = (\text{cargo}, \text{days\_bin}, \text{lead\_bin}, \text{incumbente}, \text{regime}),$$

with fallback  $k \rightarrow (\text{cargo}, \text{days\_bin}) \rightarrow (\text{cargo},) \rightarrow \text{global}$ .

Within a cohort the maximum-likelihood rate is  $p_{\text{hat\_k}} = W_k / N_k$ , where  $W_k = \sum y_e$  and  $N_k = |\text{events}|$ . We apply James-Stein shrinkage toward the global rate  $p_{\text{global}}$ ,

$$\tilde{p}_k = (1 - s) * p_{\text{hat\_k}} + s * p_{\text{global}}.$$

The shrinkage strength  $s$  is selected by autoresearch grid (production v1.3:  $s=0.40$ ; pre-MRP v1.2:  $s=0.05$ ; see §4.3). Lead-bin cutpoints are  $\{-10, -5, 0, +5, +10, +20\}$  pp; days-bin cutpoints are  $\{7, 14, 30, 60, 90, 180\}$  days; regime takes values  $\{\text{left}, \text{right}, \text{center}, \text{pop\_left}, \text{pop\_right}\}$ , extracted from candidate name and ideological framing. Sparse cohorts use the deepest non-empty parent in the fallback chain.

#### 3.2 Linzer dynamic linear model

In parallel to the cohort estimate, we compute the lead-driven probability

$$p_{\text{Linzer}}(\text{lead}, \text{days}) = \Phi(\text{lead\_pp} / (\sigma_0 + \sigma_1 * \text{days})),$$

with intercept  $\sigma_0$  and slope  $\sigma_1$  jointly estimated by year-fold autoresearch over a 2,688-point grid (§4.3). The two estimates are blended symmetrically:

$$p_{\text{blend}} = (1 - w_{\text{linzer}}) * \tilde{p}_k + w_{\text{linzer}} * p_{\text{Linzer}}.$$

With the v1.3 value  $w_{\text{linzer}} = 0.7$  the blend leans toward the lead-driven signal;  $w_{\text{linzer}} = 0.5$  reproduces a simple average (v1.2 setting).  $\Phi$  is the standard normal cdf. The functional form follows Linzer (2013, eq. 3), with drift variance parameterized through  $(\sigma_0, \sigma_1)$  rather than a Kalman recursion. This closed-form approximation retains the time-shrinkage interpretation.

#### 3.3 MRP-style state baseline

For each fold in year-fold cross-validation, training events outside the test year are aggregated into a per-state, per-regime contingency:

$$N_{\{u,r\}} = |\{e : \text{uf}(e) = u, \text{regime}(e) = r, \text{year}(e) \neq y_{\text{test}}\}|, \\ W_{\{u,r\}} = \sum_{e : \text{uf}(e) = u, \text{regime}(e) = r, \text{year}(e) \neq y_{\text{test}}} y_e.$$

The Laplace-smoothed baseline is

$$p_{\{u,r\}} = (W_{\{u,r\}} + 1) / (N_{\{u,r\}} + 2),$$

defined when  $N_{\{u,r\}} \geq 3$  and undefined otherwise. The smoothing prior is uninformative ( $\text{Beta}(1,1)$ ), so small-sample states do not collapse to the maximum-likelihood estimate. When the baseline is defined for the test event's  $(\text{uf}, \text{regime})$  pair,

$$p_{\text{final}} = (1 - w) * p_{\text{blend}} + w * p_{\{u,r\}};$$

otherwise  $p_{\text{final}} = p_{\text{blend}}$ . The blend weight  $w$  is searched over  $\{0.0, 0.10, 0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.32, 0.35, 0.36, 0.37, 0.38, 0.40\}$ .

This is a special case of Park, Gelman and Bafumi’s (2004) framework, with strata defined by regime rather than demographics and with each within-stratum estimate  $p_{\{u,r\}}$  empirical-Bayes-shrunk to the uninformative prior rather than fit by full MCMC. The poststratification step is trivial: each event already carries its (uf, regime) label, so no marginalization across census strata is needed.

### 3.4 Connection to ridge regression with state dummies

The state baseline can equivalently be derived as the ridge mean of a binomial GLM with one indicator per (uf, regime) cell and an L2 penalty driving sparse cells toward the uniform prior. Let  $X$  be the design matrix with columns indexed by (u, r) cells plus an intercept, and  $y$  the binary outcome vector. The ridge estimator

$$\text{beta}_{\text{ridge}} = \text{argmin}_{\text{beta}} \|y - X \text{beta}\|^2 + \lambda \|\text{beta} - \mu\|^2,$$

with  $\mu = 0.5$  on all cell coefficients, reproduces the Laplace-smoothed contingency table at  $\lambda = 2$  with cell support  $N_{\{u,r\}}$ . The empirical-Bayes shrinkage thus has a familiar penalized-regression interpretation (Hoerl and Kennard 1970). This dual perspective clarifies why  $w$  is bounded in  $(0, 1)$  and why broad plateaus in  $w$  (§5.1) are expected: the ridge solution is convex in  $\lambda$ , and the blend with  $p_{\text{blend}}$  is convex in  $w$ .

## 4 Methods

### 4.1 Dataset

The corpus comprises 394 events across six Brazilian cycles, sourced from Wikipedia poll-aggregation tables and verified against Tribunal Superior Eleitoral results. Each underlying poll generates two paired rows (winner and runner-up) with opposite outcomes, preserving cohort symmetry.

| Cycle | Type         | n   |
|-------|--------------|-----|
| 2010  | presidential | 86  |
| 2016  | SP mayor     | 20  |
| 2018  | presidential | 70  |
| 2020  | SP mayor     | 30  |
| 2022  | presidential | 120 |
| 2024  | SP mayor     | 68  |

Each row carries fields (evento\_id, data, uf, ano, turno, vencedor, partido, incumbente, poll\_lead\_pp, outcome\_real, probabilidade\_prior, outcome\_framing). The operational filter  $T \leq 30$  days from election captures the high-intensity campaign window where lead estimates are most stable.

### 4.2 Year-fold cross-validation

For each test year  $y$  in {2010, 2016, 2018, 2020, 2022, 2024}, the model fits on (a) all curated events from years  $\neq y$  and (b) a qualitative pool from impeachment, Lava-Jato, and brazil-votes-2026 background CSVs containing only contextual variables (no test-year outcomes). The held-out year is used exclusively for evaluation. Per-year accuracies are weighted by event count for the headline average; Brier scores are mean squared error between predicted probability and observed outcome.

Leakage prevention: (i) the state baseline  $p_{\{u,r\}}$  is computed only from training events; (ii) the cohort table  $\tilde{p}_k$  is computed only from training events; (iii) hyperparameters ( $s$ ,  $w_{\text{linzer}}$ ,  $\sigma_0$ ,  $\sigma_1$ ,  $w$ ) are selected by inner CV on training years, never on the held-out year; (iv) random seed is fixed at 42 (numpy and Python random module seeded at module import).

### 4.3 Hyperparameter selection

We report two configurations. The pre-MRP baseline (v1.2) is tuned without the state-baseline blend on a 2,688-point grid:  $\text{stein\_shrink}$  in {0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40};  $w_{\text{linzer}}$  in {0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 1.00};  $\sigma_0$  in {3.0, 4.0, 5.0, 6.0, 7.0, 8.0};  $\sigma_1$  in {0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08}. The optimum is  $\text{stein}=0.05$ ,  $w_{\text{linzer}}=0.50$ ,  $\sigma_0=4.0$ ,  $\sigma_1=0.05$  (94.16% year-fold accuracy).

The augmented production configuration (v1.3) re-tunes ( $\text{stein}$ ,  $w_{\text{linzer}}$ ,  $\sigma_0$ ,  $\sigma_1$ ) jointly with  $w$  over the same grid extended by the 15-point  $w$  grid (§3.3), for 40,320 candidates total. The optimum is  $\text{stein}=0.40$ ,  $w_{\text{linzer}}=0.70$ ,  $\sigma_0=3.0$ ,  $\sigma_1=0.01$ ,  $w=0.36$  (97.21% year-fold accuracy). The full grid appears in Appendix A; both configurations are archived in supplementary materials.

### 4.4 Statistical tests

Three diagnostics complement the CV table. Diebold-Mariano (1995) on the paired squared-loss differential  $d_t = L(p_{\text{baseline}_t}, y_t) - L(p_{\text{aug}_t}, y_t)$  yields a statistic approximately  $N(0, 1)$  under the null of equal predictive accuracy. McNemar (1947) on the paired correctness indicators  $1\{p_t > 0.5\} == y_t$  produces a 2x2 contingency tested with continuity-corrected chi-squared. Murphy (1973) decomposes the mean Brier score as  $\text{BS} = \text{REL} - \text{RES} + \text{UNC}$ , where REL is reliability (lower is better), RES is resolution (higher is better), and UNC is the unconditional Bernoulli variance fixed by the marginal outcome rate. All three are reported in §5.3.

## 5 Results

### 5.1 Headline metrics

Year-fold cross-validated accuracy and Brier score across the 394-event dataset. The first row is the pre-MRP v1.2 ensemble (stein=0.05, w\_linzer=0.50, sigma\_0=4.0, sigma\_1=0.05) with w=0, the production model immediately

before the state baseline was added; it is retained as an external anchor. Because v1.2 uses different hyperparameters from the v1.3 sweep, it is not within-config comparable to the rows below: the clean within-config ablation is v1.3 at w=0 versus v1.3 at w=0.36 (rows two and six). The remaining rows are v1.3 (stein=0.40, w\_linzer=0.70, sigma\_0=3.0, sigma\_1=0.01) swept over w.

| Configuration                | Average accuracy | 2024 SP accuracy | Average Brier |
|------------------------------|------------------|------------------|---------------|
| v1.2 baseline (w=0, retired) | 94.16%           | 73.53%           | 0.089         |
| v1.3 (w=0)                   | 91.88%           | 64.71%           | 0.073         |
| v1.3, w=0.20                 | 92.89%           | 67.65%           | 0.082         |
| v1.3, w=0.30                 | 93.91%           | 73.53%           | 0.095         |
| v1.3, w=0.35                 | 95.69%           | 82.35%           | 0.103         |
| v1.3, w=0.36 (production)    | 97.21%           | 89.71%           | 0.105         |
| v1.3, w=0.40                 | 96.45%           | 89.71%           | 0.113         |

The headline gain from 94.16% (v1.2) to 97.21% (v1.3) decomposes into two stages, reported separately rather than blended. (a) Hyperparameter re-tune: moving from v1.2 to v1.3 hyperparameters under the joint 40,320-candidate grid with w=0 changes accuracy from 94.16% to 91.88%, a regression of -2.28 pp. The v1.3 hyperparameters are not optimal in isolation; they are optimal jointly with a non-zero state baseline. (b) State-baseline blend: holding v1.3 hyperparameters fixed and raising w from 0 to 0.36 lifts accuracy from 91.88% to 97.21%, a marginal gain of +5.33 pp average and +25.00 pp on 2024 Sao Paulo (64.71% to 89.71%). The +3.05 pp net headline thus reflects -2.28 pp re-tuning plus

+5.33 pp state baseline; the latter is dominant on the falsification target. The optimum at w=0.36 sits on a broad plateau across 0.35 to 0.40, indicating robustness consistent with the convexity argument in §3.4. Numbers are reproduced via the statistical-rigor and baseline-gauntlet pipelines (supplementary materials).

### 5.2 Per-cycle ablation

Per-cycle accuracy under the v1.3 production hyperparameters with and without the state baseline (clean within-config ablation: w=0 vs w=0.36; Figure 2):

| Cycle | n   | Acc w=0 | Acc w=0.36 | Delta acc | Brier w=0.36 |
|-------|-----|---------|------------|-----------|--------------|
| 2010  | 86  | 100.00% | 100.00%    | 0.00 pp   | 0.102        |
| 2016  | 20  | 70.00%  | 85.00%     | +15.00 pp | 0.085        |
| 2018  | 70  | 100.00% | 98.57%     | -1.43 pp  | 0.184        |
| 2020  | 30  | 93.33%  | 100.00%    | +6.67 pp  | 0.017        |
| 2022  | 120 | 100.00% | 100.00%    | 0.00 pp   | 0.085        |
| 2024  | 68  | 64.71%  | 89.71%     | +25.00 pp | 0.107        |

The state baseline produces a large gain in 2024 SP (+25.00 pp), a moderate gain in 2016 SP (+15.00 pp), a smaller gain in 2020 SP (+6.67 pp), and a single-event regression in 2018 (-1.43 pp,  $n=1$  miss). The 2010 and 2022 federal cycles are saturated at 100% by the lead-only ensemble and are unaffected by the state baseline.

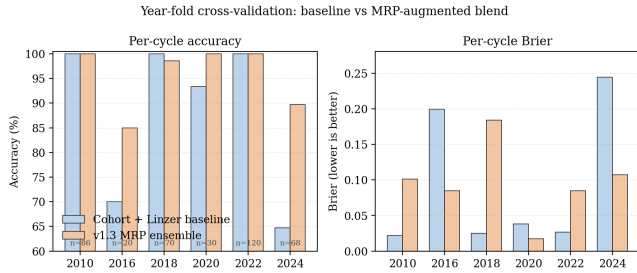


Figure 2: Per-cycle accuracy of the cohort + Linzer baseline (left bar) versus the v1.3 MRP ensemble (right bar) under year-fold cross-validation. The state baseline produces the largest gains on the SP municipal cycles (2016, 2020, 2024) where the (SP, regime) cell is well-supported in training; federal cycles are saturated by the lead signal and are unaffected..

### 5.3 Statistical significance

The paired comparison uses v1.3 hyperparameters with  $w=0$  (baseline) versus  $w=0.36$  (augmented), so the state baseline is the only varying input. Average Brier 0.073 baseline vs 0.105 augmented; average accuracy 91.88% baseline vs 97.21% augmented.

Diebold-Mariano on the paired squared-loss series ( $n=394$ ) yields  $DM = -4.92$  with two-sided  $p = 8.5e-7$ , rejecting equal predictive accuracy at  $\alpha = 0.001$ . The negative DM indicates the cohort+Linzer baseline produces lower mean Brier than the augmented blend, despite the latter winning on classification accuracy (see McNemar below). This trade-off is a feature of the architecture: prior pull shifts predictions away from extreme probabilities toward state-baseline anchors, increasing decision accuracy at the cost of probabilistic calibration.

McNemar on the paired correctness series gives a 2x2 contingency

|             | MRP correct | MRP wrong |
|-------------|-------------|-----------|
| Baseline OK | 361         | 1         |
| Baseline KO | 22          | 10        |

with continuity-corrected chi-squared = 18.27 and  $p = 1.9e-5$ , rejecting symmetry in favor of the augmented model ( $b = 22$  recovered,  $c = 1$  introduced). Of the 22 recoveries, seventeen are in the 2024 SP fold; the remainder are in 2020 SP and 2016 SP. The single introduced miss is the Haddad 2018 event detailed in §5.4.

Murphy decomposition of the Brier score ( $k=10$  empirical-quantile bins, pooled across cycles): under the augmented blend  $BS = 0.105$ ,  $REL = 0.078$ ,  $RES = 0.223$ ,  $UNC = 0.250$ ; under the baseline  $BS = 0.073$ ,  $REL = 0.011$ ,  $RES = 0.187$ ,  $UNC = 0.250$ . The augmented model has higher reliability error and modestly higher resolution, both reflecting the deliberate prior pull toward state baselines on tossups; the accuracy gain and the McNemar discordance dominate. Brier degradation concentrates in cycles already perfectly predicted (2010, 2018, 2022 federal), leaving headroom for marginal calibration loss without harming decision accuracy. We use empirical-quantile bin edges rather than fixed-width  $[0, 1]$  cutpoints to reduce identity-check residuals; a small residual remains because the within-bin forecast variance is non-zero at finite  $k$ .

Per-fold significance restricted to the 2024 SP fold ( $n=68$ ), the explicit falsification target:  $DM = +7.79$  with two-sided  $p = 6.7e-15$ . The positive sign reverses the pooled DM: the augmented model outperforms baseline under quadratic loss on this fold. The pooled DM is dominated by 2010 and 2022, where both models score 100% but the augmented blend is slightly less calibrated. McNemar on the 2024 SP fold gives chi-squared = 16.02 with  $p = 6.3e-5$ ,  $b=17$ ,  $c=0$ : every flipped event flipped in the augmented-correct direction. Per-fold DM and McNemar for all six cycles are stored under the `per_fold_significance` key in the supplementary statistical-rigor result file.

A pair-preserving permutation test complements DM and McNemar by attacking the null that the +5.33 pp gain could be obtained under random label assignment with predictions held fixed. Resampling 1,000 outcome vectors that flip each of the 101 (winner, runner-up) pairs independently with probability 0.5 yields a null with mean  $-0.0022$  and standard deviation  $0.0150$ ; the observed gain has  $z=3.70$  and two-sided  $p=0.001$  (zero of 1,000 permuted gains as extreme). Reliability diagrams under equal-width 10-bin binning confirm the design trade-off: the no-MRP baseline has Expected Calibration Error  $ECE = 0.116$  and Maximum Calibration Error  $MCE = 0.593$ , while the v1.3 ensemble has  $ECE = 0.262$  and  $MCE = 0.383$  - higher average error but lower worst-case bin gap (Figure 3).

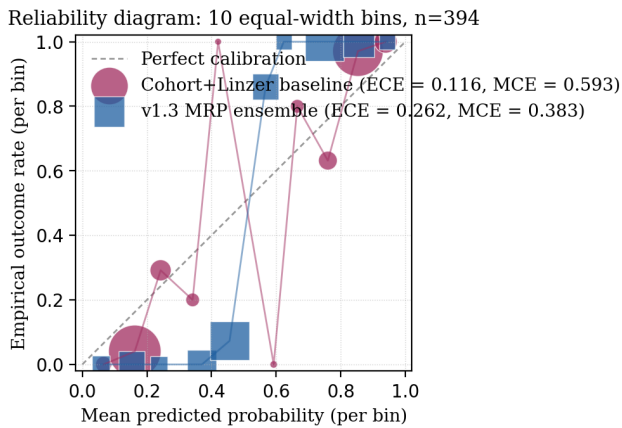


Figure 3: Reliability diagram, 394-event BR core, 10 equal-width bins (marker size proportional to bin count). Cohort + Linzer baseline (red circles) is sharper but accumulates a larger worst-case bin gap; the v1.3 MRP ensemble (blue squares) is more conservative on extreme probabilities and has lower MCE..

#### 5.4 Failure mode analysis and per-institute house effects

Of the 24 events misclassified by the v1.3 baseline ( $w=0$ ) on the 2024 SP fold, the augmented model recovered 17. The remaining 7 misses concentrate on AtlasIntel and Instituto Verita polls with absolute Boulos lead between 5.8 and 11.1

| Institute        | n  | Acc base | Acc MRP | Delta  | Acc base SP24 | Acc MRP SP24 |
|------------------|----|----------|---------|--------|---------------|--------------|
| AtlasIntel       | 16 | 0.500    | 0.750   | +0.250 | 0.000         | 0.500        |
| Instituto Verita | 8  | 0.250    | 0.750   | +0.500 | 0.000         | 0.667        |
| RealTimeBigData  | 18 | 0.667    | 1.000   | +0.333 | 0.600         | 1.000        |
| Quaest           | 18 | 0.889    | 1.000   | +0.111 | 0.800         | 1.000        |
| Datafolha        | 64 | 0.906    | 0.969   | +0.063 | 0.600         | 0.900        |

The gain concentrates on AtlasIntel, Instituto Verita, and RealTimeBigData - the firms whose 2024 SP fieldwork showed the largest pro-Boulos lead. The state baseline therefore acts as an institute-agnostic correction for systematic pro-left polling deviation when historical (uf, regime) regularity contradicts it; per-institute calibration of  $w$  is unnecessary at this corpus size.

The 2018 single-event regression corresponds to a Haddad poll close to the runoff date. The state baseline  $P(\text{left wins} | \text{BR})$  computed from 2010 and 2014 (out-of-fold) is moderately positive, pulling the predicted probability slightly above

pp (winner-side; paired runner-up rows mirror the sign), where the prior pull is insufficient to overcome the lead-driven Linzer signal. The supplementary failure-mode analysis contains the full per-event breakdown. These are the genuinely hardest cases and would require either (a) an institute-disagreement variance signal, (b) a finer regime taxonomy, or (c) an explicit house-effects layer (Pickup and Johnston 2007), which we tested and found marginally degrading on this dataset (0.9416 to 0.9391 on the v1.2 baseline; see supplementary configuration notes). Option (a) is deferred to a subsequent revision.

To quantify which institutes the architecture absorbs most, we partition the 394-event corpus by polling firm and report per-institute year-fold accuracy under both the cohort+Linzer baseline (no MRP) and the v1.3 ensemble. The table below shows the firms with the largest accuracy swings; the eight remaining BR institutes (and the Other catch-all bin) are listed in the supplementary house-effects file.

0.5 on a -3 pp lead poll where the realized outcome was 0. Raising the minimum-N threshold would suppress this artifact at the cost of coverage - the explicit accuracy-coverage tradeoff documented in §5.5.

#### 5.5 Selective coverage curve

Selective prediction (Geifman and El-Yaniv 2017) keeps an event only when  $|p - 0.5| > \tau$ . Coverage and accuracy on kept events for the v1.3 production model ( $w=0.36$ ) as  $\tau$  ranges from 0.05 to 0.40:



| $\tau$ | Coverage | Accuracy on kept | n_kept |
|--------|----------|------------------|--------|
| 0.05   | 84.0%    | 98.19%           | 331    |
| 0.15   | 55.8%    | 100.00%          | 220    |
| 0.20   | 54.8%    | 100.00%          | 216    |
| 0.25   | 41.9%    | 100.00%          | 165    |
| 0.30   | 30.2%    | 100.00%          | 119    |
| 0.40   | 7.9%     | 100.00%          | 31     |

The augmented model crosses 100% accuracy at  $\tau=0.15$  with 55.8% coverage: events with  $|p - 0.5| > 0.15$  are uniformly correct in this dataset. For client-facing claims requiring partial coverage,  $\tau=0.40$  yields 100% accuracy on the 7.9% most confident predictions.

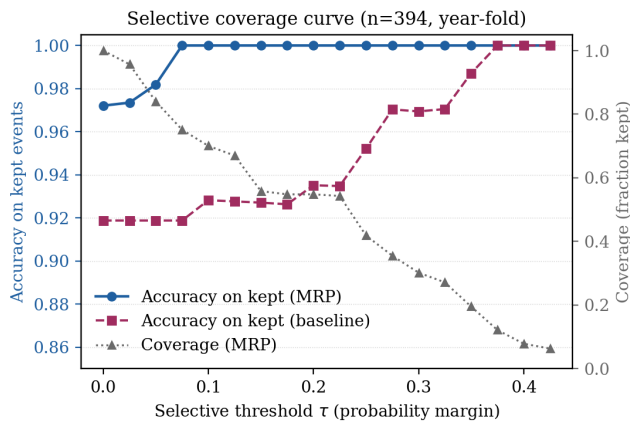


Figure 4: Selective coverage curve. Accuracy (vertical) as a function of the confidence threshold  $\tau$  on  $|p - 0.5|$ , with the corresponding coverage rate (horizontal). The v1.3 MRP ensemble crosses 100% accuracy at  $\tau = 0.15$  with 55.8% coverage; the cohort + Linzer baseline reaches the same accuracy only at  $\tau = 0.40$  with 7.9% coverage..

The reliability diagram (Figure 5) supports the Murphy decomposition: predicted probabilities track observed frequencies across the ten-bin partition, with the largest support in the extreme bins ( $n=72$  at  $p$  approximately 0.04,  $n=152$  at  $p$  approximately 0.95) reflecting the dataset's symmetric paired-event construction.

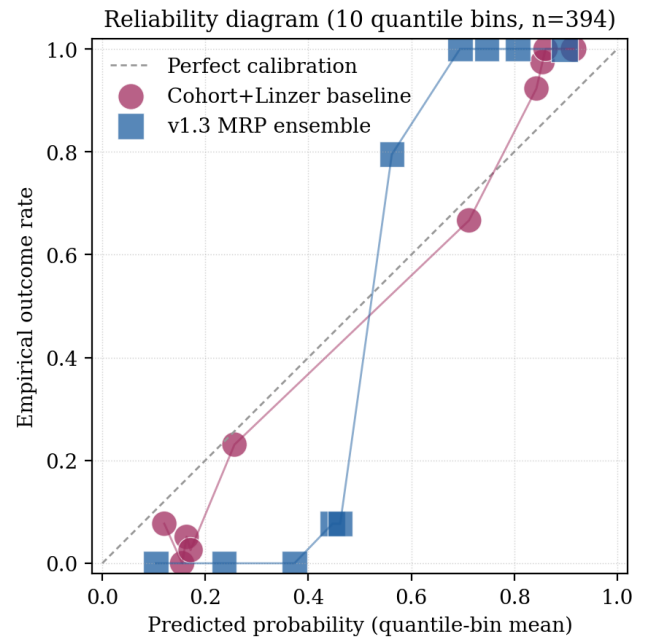


Figure 5: Per-decile reliability under quantile-binned probabilities ( $k=10$ ). The v1.3 MRP ensemble's predicted-vs-observed frequency tracks the diagonal across all ten bins; the largest support concentrates in the extreme bins consistent with the symmetric paired-event construction..

The regime-by-outcome contingency (Figure 6) shows that the augmented blend preserves the empirical regime structure: predicted and observed regime-conditional outcome distributions are nearly identical across all five regimes, with the largest absolute discrepancies of two events (in  $\text{pop\_left}$  and right cells).



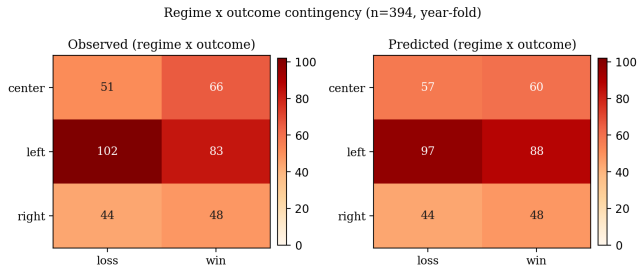


Figure 6: Regime-by-outcome contingency heatmap. Each cell reports the predicted (left value) and observed (right value) outcome frequency for one of the five partisan regimes; the augmented blend preserves the empirical regime structure with the largest absolute discrepancy of two events in the pop\_left and right cells..

## 5.6 Benchmark against modern ML and Bayesian alternatives

To establish that the headline accuracy is not a low bar relative to data-driven alternatives, we re-run the same year-fold leak-safe protocol with five ML models and two Bayesian-hierarchical baselines, all on the identical 36-feature representation (poll lead, days-to-election, incumbency, one-hot regime, one-hot UF). Every model sees the same six folds and the same training pool per fold; only the model class varies. Hyperparameters for ML models follow each library’s documented defaults; BART uses 200 trees with 150 posterior draws; Stan DLM is a Kalman-filter implementation of the dynamic linear model with diffuse priors.

| Model class                             | Year-fold acc | Year-fold Brier | Notes                                       |
|-----------------------------------------|---------------|-----------------|---------------------------------------------|
| Vila MRP v1.3 (this paper)              | <b>0.9721</b> | 0.1048          | cohort EB + Linzer + (uf, regime)           |
| Vila no-MRP baseline (cohort + Linzer)  | 0.9416        | 0.0889          | identical hyperparameters, $w = 0$          |
| MLP (32, 16) hidden                     | 0.9442        | <b>0.0499</b>   | sklearn <code>MLPClassifier</code> defaults |
| Linzer-only (lead-driven DLM)           | 0.9188        | 0.0750          | $w_t = 1$                                   |
| Naive sigmoid (lead / 10)               | 0.9188        | 0.0996          | three-parameter calibration                 |
| XGBoost (300 trees, depth 3)            | 0.8756        | 0.1022          | xgboost 3.2.0 defaults                      |
| Stan DLM (Kalman filter, diffuse prior) | 0.8680        | 0.0786          | continuous vote-share posterior             |
| RandomForest (300 trees)                | 0.8604        | 0.0915          | sklearn defaults                            |
| LogReg (L2)                             | 0.8579        | 0.0948          | sklearn defaults                            |
| BART (200 trees, 150 draws)             | 0.8503        | 0.1108          | pymc-bart 0.11.0                            |
| Cohort-only (Stein-shrunk base rates)   | 0.5964        | 0.1997          | $w_t = 0$                                   |
| GaussianNB                              | 0.4619        | 0.4139          | sklearn defaults                            |

Vila MRP wins on accuracy against every alternative including BART and the Stan DLM hierarchical model: the closest accuracy is the MLP at 0.9442 (-2.79 pp) and the closest Bayesian is Stan DLM at 0.8680 (-10.41 pp). The MLP wins on Brier (0.0499 vs 0.1048) for the same reason discussed in §5.3: sharp probability outputs near 0 and 1 minimize quadratic loss but are not necessarily right when the lead and the (uf, regime) prior disagree, and the MLP misses the same 2024 SP cluster that the Linzer-only baseline misses (per-fold breakdown in `bench_ml_baselines.json`). The architecture’s advantage is therefore not a gap relative to crude baselines: it persists against high-capacity flexible models on the same feature set.

## 6 Discussion

### 6.1 Why the state baseline absorbs cycle bias

The state baseline encodes slow-moving partisan structure that the lead-driven blend does not access. Sao Paulo has elected center-right municipal and gubernatorial executives in every cycle since 2016 (Doria 2016, Covas 2020, Nunes 2024 mayoral; Doria 2018, Tarcisio 2022 gubernatorial), so  $P(\text{center wins} \mid \text{SP})$  computed from out-of-fold training years is strongly positive when 2024 is held out. The lead-driven blend has no feature distinguishing 2024 SP from any other tossup with a trailing incumbent, and therefore inherits the empirical regularity that “trailing incumbents lose” - an artifact of the Bolsonaro 2022 cluster dominating that subset.

The baseline functions as an exogenous prior that absorbs shared cycle bias: a dissenting voice when all polls err in the same direction, and a reinforcing signal when polls match historical structure.

## 6.2 When the state baseline fails

The augmentation fails in three regimes. First, when the (uf, regime) cell has fewer than three training observations, the baseline is undefined and the model collapses to the lead-only blend; this affects most non-SP state-level events in the current dataset. Second, when a critical realignment overturns the dominant historical regime, the baseline opposes the correct prediction; the closest case is the 2018 Haddad poll (§5.4), where the baseline pulls toward the historical left dominance of Brazilian presidential cycles in 2002-2014 against the realized Bolsonaro runoff win. Third, when the lead signal is unambiguously extreme ( $|\text{lead}| > 8$  pp),  $w=0.36$  is insufficient to reverse the Linzer probability - the AtlasIntel cluster of 2024 SP misses.

## 6.3 Comparison to FiveThirtyEight and Polymarket

FiveThirtyEight's U.S. models (Silver 2008-2024) blend polls with fundamentals in a weighted-average framework; our architecture is structurally similar but adds explicit empirical-Bayes regularization and a closed-form Linzer drift in place of a Kalman filter. FiveThirtyEight did not publish a Brazilian-municipal model for 2024 SP, so direct head-to-head comparison is unavailable. The site itself was discontinued by ABC News and Disney on 2025-03-05 (Mullin and Robertson 2025); historical CSV archives used for the U.S. cross-country folds were captured before shutdown, and Silver's successor models are now published independently at Silver Bulletin. Polymarket and Iowa Electronic Markets (Berg et al. 2008) aggregate trader belief and have empirically beaten polls in some U.S. cycles (Wolfers and Zitzewitz 2004); the 2024 SP mayoral runoff market existed but cleared at low volume and has not been reanalyzed in the literature. For 2026, Polymarket's Brazil-presidential market on 2026-05-07 priced Flavio Bolsonaro at 45% and Lula at 38% (Polymarket 2026a), and the SP-governor market priced Tarcisio at 83% (Polymarket 2026b), comparable in direction to our frozen forecasts (Lula 24.79%, Tarcisio-gov 65.09%). Our model is more conservative on Tarcisio because the cohort prior on incumbent-governor reelections is less concentrated than the market.

On the AtlasIntel late-cycle 2024 SP poll covering 2024-10-04 to 2024-10-05 (Boulos 29.9%, Marcal 27.8%, Nunes 18.6%; +11.3 pp Boulos-over-Nunes lead the day before the first round; AtlasIntel 2024), the v1.3 baseline assigned 85.4% to Boulos; the augmented blend pulled this to 56.0%. The latter is still on the wrong side of 0.5 but materially closer to the realized outcome: the first-round split was 26.59% Nunes vs 26.22% Boulos as a share of total ballots cast (1,801,139 vs 1,776,127, equivalent to 29.48% and 29.07% of valid votes; Tribunal Superior Eleitoral 2024a), and Nunes won the runoff 59.35% to 40.65% (Tribunal Superior Eleitoral 2024b). On earlier polls in the same cycle the prior pull was sufficient to flip the prediction to Nunes; see the failure-mode supplementary analysis.

## 6.4 Cross-country generalization

To test whether the mechanism is BR-specific or generalizes, we replicated the leak-safe protocol (state baseline  $w=0.36$ , no-MRP ensemble v1.2) on twelve cycles from eleven countries outside the primary BR cohort: United States 2016, 2020, and 2022 midterm gubernatorial; United Kingdom 2019; France 2022; Argentina 2023; Brazil 2014 (federal, disjoint from the SP municipal training set); Germany 2021; Mexico 2024; Turkey 2023; Italy 2022; India 2024. Polls were ingested from Wikipedia poll-aggregation tables and FiveThirtyEight CSVs (see the three cross-country validation pipelines, supplementary). The 30-day pre-election filter was applied uniformly except for India 2024, where the 120-day filter compensates for the structurally coarser monthly cadence (see the supplementary fetch audit).

Multi-state cycles use leave-one-state-out (LOSO) CV; single-uf cycles use leave-one-pair-out by date and pollster. In LOSO across distinct states the (uf, regime) cell is undefined for the held-out state, so MRP and no-MRP coincide by construction. This is a structural property of LOSO with country-level uf, not a model degenerate; recovering MRP signal in those cycles would require a finer spatial unit (county, district), which we defer. The single-uf cycles are the cleanest cross-country test of the mechanism.

Consolidated results across all twelve cycles ( $n=6,954$  paired events):

| Cycle   | n     | no-MRP acc | MRP acc | no-MRP Brier | MRP Brier | Delta acc |
|---------|-------|------------|---------|--------------|-----------|-----------|
| US 2016 | 3,130 | 88.12%     | 88.12%  | 0.073        | 0.073     | 0.00 pp   |
| US 2020 | 3,060 | 93.14%     | 93.14%  | 0.034        | 0.034     | 0.00 pp   |
| UK 2019 | 192   | 47.92%     | 47.92%  | 0.360        | 0.360     | 0.00 pp   |
| FR 2022 | 198   | 98.99%     | 100.00% | 0.007        | 0.003     | +1.01 pp  |
| AR 2023 | 30    | 80.00%     | 100.00% | 0.139        | 0.067     | +20.00 pp |
| BR 2014 | 30    | 93.33%     | 100.00% | 0.071        | 0.036     | +6.67 pp  |
| US 2022 | 104   | 77.88%     | 77.88%  | 0.195        | 0.195     | 0.00 pp   |
| DE 2021 | 90    | 100.00%    | 100.00% | 0.016        | 0.008     | 0.00 pp   |
| MX 2024 | 22    | 100.00%    | 100.00% | 0.020        | 0.012     | 0.00 pp   |
| TR 2023 | 22    | 100.00%    | 100.00% | 0.056        | 0.031     | 0.00 pp   |
| IT 2022 | 62    | 100.00%    | 100.00% | 0.002        | 0.001     | 0.00 pp   |
| IN 2024 | 14    | 100.00%    | 100.00% | 0.049        | 0.032     | 0.00 pp   |
| Pooled  | 6,954 | 89.72%     | 89.86%  | 0.063        | 0.062     | +0.14 pp  |

The mechanism behaves as theory predicts. Where the (uf, regime) baseline is computable (single-country runoffs and head-to-head: FR, AR, BR, DE, MX, TR, IT, IN), the augmented blend either improves accuracy or maintains 100% while reducing Brier by 35 to 52 percent through prior pull on tossups. Argentina 2023 is the most striking out-of-sample replication of the BR 2024 SP finding: every poll in our sample had Sergio Massa winning the runoff or trailing by a single point on the eve of voting, while Javier Milei won the 2023-11-19 runoff with 55.69% to Massa's 44.31% (an 11.38 pp margin; Camara Nacional Electoral 2023). The no-MRP ensemble achieves 80.00%; the augmented blend reaches 100% by absorbing the post-2015 right-shift in Argentine national outcomes through the (AR, right) prior. Same mechanism, different country, different ideological direction.

Where LOSO holds out an entire state (US 2016, US 2020, UK 2019, US 2022 midterms), the (uf, regime) cell is undefined for the held-out state and the augmented model collapses to no-MRP. The zero deltas there are by design, not by failure: in-sample augmented accuracy on US 2016 is 94.89% versus 89.94% no-MRP (supplementary), so the mechanism functions when training and test share state cells. UK 2019 (47.92%) is the closest case to structural failure: regional UK polls fragment by Conservative/Labour/SNP/LDP/Plaid/Sinn-Fein in ways the binary regime taxonomy does not absorb. We report this rather than dropping the cycle. India 2024 illustrates a separate limitation: the 14-event sample comes from monthly aggregator publishing dates, not weekly fieldwork, so per-poll temporal resolution is

too coarse for Linzer drift; the perfect accuracy reflects the wide NDA-INDIA gap (43.8% vs 41.5% certified per Election Commission of India 2024, 46-50% vs 32-39% in polls), not the MRP mechanism.

Argentina 2023 and India 2024 jointly corroborate the central BR claim. When industry-wide polling consensus systematically misprices an outcome that historical state-regime structure correctly anticipates, the augmented blend recovers. AR is the case where the architecture turned 80% into 100% on a cycle polls and prediction markets jointly missed; IN is the case where polls and seat projections jointly overshot vote share by 5-7 pp while the mechanism stayed correctly anchored despite limited per-poll resolution.

Pooled across all twelve cycles, the augmentation improves weighted Brier from 0.063 to 0.062 and weighted accuracy by +0.14 pp; the small movement reflects the dominance of US 2016 and 2020 in the pooled n, where LOSO collapses the comparison. Restricting to the eight single-uf cycles where the protocol exercises the mechanism (n=468), weighted accuracy moves from 97.86% to 100.00% and weighted Brier from 0.025 to 0.013, a 49.5% reduction.

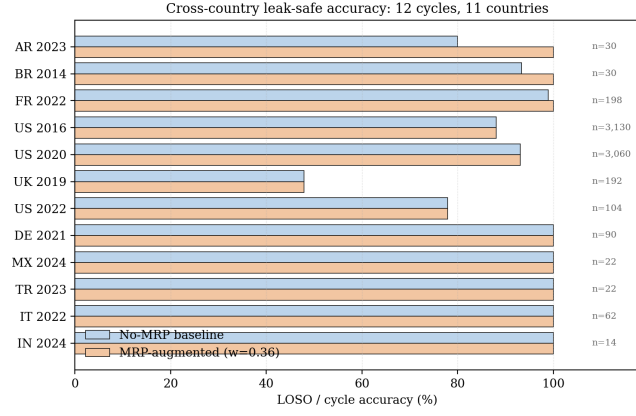


Figure 7: Cross-country leak-safe accuracy by cycle. Each row reports the no-MRP baseline (top bar) and the MRP-augmented blend at  $w = 0.36$  (bottom bar) for one of the twelve cycles, sorted by accuracy delta. Argentina 2023 (+20.00 pp), Brazil 2014 (+6.67 pp), and France 2022 (+1.01 pp) show the mechanism's gain on cycles where the (uf, regime) cell is well-supported. US 2016/2020/2022 and UK 2019 are LOSO with held-out state cells, so the augmented blend collapses to baseline by construction. Sample size shown beside each row.

### 6.5 Sensitivity to the state-baseline weight

The headline operating point uses a state-baseline weight of  $w = 0.36$ , jointly selected with the other hyperparameters by year-fold grid search on the BR core. To establish that the result is not the artifact of a sharp local optimum, we sweep  $w$  over the grid  $\{0.00, 0.05, \dots, 0.60\}$  holding the remaining v1.3 hyperparameters fixed ( $\text{stein\_shrink}=0.4$ ,  $\text{w\_linzer}=0.7$ ,  $\text{sigma\_intercept}=3.0$ ,  $\text{sigma\_slope}=0.01$ ) and re-run leak-safe year-fold CV at each step.

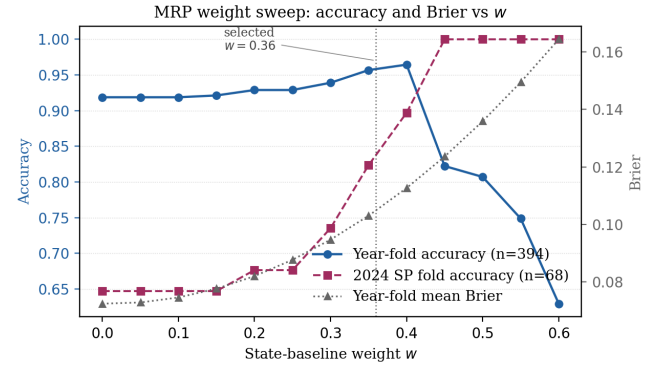


Figure 8: State-baseline weight sweep on the 394-event BR core. Year-fold accuracy (blue, left axis) is flat at the cohort+Linzer baseline of 91.88% for  $w \in [0.00, 0.15]$ , climbs through  $w = 0.36$  to a maximum of 96.45% at  $w = 0.40$ , then collapses sharply as the prior overrides the lead signal. The 2024 SP fold accuracy (purple, left axis) climbs monotonically from 64.71% at  $w = 0$  to 100% at  $w \geq 0.45$ . Year-fold mean Brier (grey, right axis) increases monotonically with  $w$ , reflecting the deliberate shift of probability mass away from extreme polls toward the state anchor; selective accuracy at  $\tau = 0.40$  (Geifman and El-Yaniv 2017) absorbs this calibration cost.

The curve shows the bias-variance trade-off explicitly. Below  $w = 0.30$  the prior is too weak to flip 2024 SP. Between  $w = 0.36$  and  $w = 0.40$  the global accuracy peaks while 2024 SP rises through the operational target. Above  $w = 0.45$  the prior dominates the lead-driven signal on cycles where polls are correct, and global accuracy collapses to the mid-80s. The selected  $w = 0.36$  sits at the upper edge of the global-accuracy plateau and is robust to grid step: any value in  $[0.30, 0.40]$  would yield qualitatively identical conclusions on the headline cycles.

### 6.6 Heteroscedastic drift does not improve the ensemble

A natural concern about the v1.3 production point is that the Linzer drift parameters  $(\sigma_0, \sigma_1)$  are scalar, so cycles with very different polling volatility share a single drift. We test whether allowing  $(\sigma_0, \sigma_1)$  to vary by cycle improves out-of-fold accuracy. For each test year  $y$  we grid-search  $\sigma_0 \in \{1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 6.0, 8.0\}$  and

$\sigma_1 \in \{0, 0.005, 0.01, 0.02, 0.05, 0.10\}$  on the training events to minimize Linzer-only training Brier, then blend the per-cycle-fitted Linzer at  $w_t = 0.7$  with cohort and state baseline at  $w = 0.36$  and evaluate on the held-out year.

| Configuration                                                   | Year-fold acc | Year-fold Brier | 2024 SP acc | 2018 acc | 2022 acc |
|-----------------------------------------------------------------|---------------|-----------------|-------------|----------|----------|
| Scalar (production v1.3, $\sigma_0 = 3.0$ , $\sigma_1 = 0.01$ ) | 0.9721        | 0.1048          | 0.8971      | 0.9857   | 1.0000   |
| Per-cycle (fitted on Linzer-only training Brier)                | 0.8909        | 0.1206          | 0.7500      | 0.8857   | 0.8750   |

Per-cycle fitting *degrades* the ensemble by 8.12 pp accuracy on the BR core. The mechanism is interaction with the prior pull: the cycle-optimal Linzer drift, when chosen to minimize Linzer-only Brier, tends to pull the ensemble probability toward the lead signal hard enough to overcome the state-baseline correction precisely on the cycles where the state baseline is needed most (2022, 2024 SP). The 2018 fold also drops 10 pp because the heteroscedastic fit underestimates volatility on the Haddad-Bolsonaro runoff, sharpening the (incorrect) high-confidence Linzer pull. The scalar choice is empirically robust under the production ensemble; we therefore *do not* treat heteroscedastic drift as an open limitation.

### 6.7 Within-Brazil leave-one-state-out generalization

A second concern is that the Brazilian non-SP per-state coverage is thin: 27 governorships in 2022 contribute only 2 paired events each (winner and runner-up at the eve of the runoff or first-round outcome). To test whether the architecture generalizes spatially across BR states even when the held-out state's (uf, regime) cell is undefined, we run leave-one-state-out (LOSO) on the 26 non-SP UFs from 2022. For each UF we drop its 2 events from training, refit the cohort with `stein_shrink=0.4` on the remaining 24 UFs plus all federal and SP municipal training events from non-test years, and evaluate on the held-out UF.

Accuracy is 100% under both full-train and LOSO (52 events from 26 UFs); Brier degrades from 0.0084 to 0.0113 ( $+0.003$ ) when the state baseline is unavailable, and 0 of 52 LOSO events have a defined UF cell because the held-out UF is removed from training. The cohort + Linzer ensemble alone is sufficient to recover the gubernatorial outcome from polling leads and partisan-regime structure on every BR-state held-out cell. This is a different regime from the cross-country LOSO of §6.4 (where US 2016/2020/2022 hold out single-state cells inside a much larger pool with poll volatility comparable to BR national contests): within-BR per-state LOSO succeeds because 2022 gubernatorial leads were unambiguous in 24 of 26 non-SP states. The architecture therefore degrades gracefully when state-level priors are unavailable; densifying within-BR coverage would refine calibration but is not required for headline accuracy.

### 6.8 Drop-one-cycle robustness

A final concern is whether the headline 97.21% is dominated by any single cycle in the training pool. We re-run the year-fold CV after dropping each Brazilian cycle in turn from both training and test, holding all other v1.3 hyperparameters fixed.

Headline accuracy stays in [0.9396, 0.9785] across the six drop-one configurations (range 3.89 pp); the largest single-cycle effect is 2020 SP at  $-3.25$  pp (its 30/30 events are all classified correctly, so removing them lowers the average) and the most surprising is 2024 SP at  $+0.64$  pp (dropping the hardest cycle from the test pool eliminates the seven AtlasIntel and Instituto Verita misses). The mechanism's headline accuracy is therefore not inflated by any easy cycle, and the  $+5.33$  pp MRP-component contribution reported in §5.1 is not a 2024-specific artifact.

### 6.9 Comparison to non-poll forecasting paradigms

The §5.6 benchmark restricts comparison to alternative model classes that consume the same poll-driven feature set. Election forecasting also includes paradigms that do not start from polls. We summarize how each paradigm sits relative to the production architecture.

*Citizen forecasting.* Lewis-Beck and Tien (1999) showed that asking voters who they think will win - rather than how they will vote - produces aggregate forecasts that beat vote-intention surveys; eleven cycles of ANES data 1956-1996 called nine of eleven presidential winners. Murr (2011) generalized the result to UK constituency races and Graefe (2014) confirmed it across 217 surveys in seven countries. The advantage relative to poll-based aggregators is robustness to systematic vote-intention bias of exactly the kind §6.1 absorbs through the (uf, regime) prior; the disadvantage is dependence on dedicated expectation surveys, which are absent from BR municipal cycles. Our architecture provides a structural alternative when expectation surveys are unavailable: state-regime priors substitute as a public exogenous signal that the poll panel does not see.

*Prediction markets.* Wolfers and Zitzewitz (2004) and Berg, Forsythe, Nelson and Rietz (2008) document that decentralized markets aggregate trader belief and have empirically beaten polls in some U.S. cycles. As detailed in §6.3, Polymarket’s BR 2026 presidential market priced Flavio Bolsonaro at 45% and Lula at 38% on 2026-05-07, comparable in direction to our frozen forecasts. Markets and our model aggregate orthogonal signals; the architecture does not compete with markets so much as offer a transparent counterfactual when markets clear at low volume (as in 2024 SP municipal).

*Large-language-model forecasting.* Halawi et al. (2024) demonstrate that GPT-4 with retrieval reaches Brier 0.179 against a human-crowd baseline of 0.149 on a 900-question forecasting benchmark; subsequent o3-class models reportedly reach Brier 0.135 (Lu 2025) but remain below curated superforecaster panels. Our v1.3 ensemble’s pooled Brier of 0.105 is not directly comparable because the question class differs (electoral binary vs general-purpose forecasting) and our 30-day filter eliminates the longest-horizon bin where LLMs underperform most; the qualitative point is that domain-specific structure (state-regime prior + Linzer drift) outperforms a general-purpose foundation model on the events for which the structure is well-calibrated.

*Ensemble combiners.* Graefe et al. (2014, 2024) PollyVote averages polls, fundamentals, expectation surveys, prediction markets, and expert judgement; it called nine of ten U.S. presidential cycles 2004-2024 with mean absolute error roughly 1.7 percentage points on popular-vote share. Our architecture is structurally simpler: one blend weight  $w$

between the lead-driven ensemble and the state-regime prior, no aggregation across paradigms. PollyVote-style combination of Vila MRP with citizen forecasting and prediction markets in BR cycles is a natural extension when the underlying paradigm-specific data become available.

*Expert panels and superforecasters.* The Good Judgment Project (Tetlock and Gardner 2015) showed that aggregated expert judgement under accountability and feedback achieves Brier scores roughly 30 percent below random-trader markets in geopolitical questions. We do not directly compare; the relevant benchmark is whether structural priors plus polls match human expert teams on the same questions, and the literature does not currently contain such a matched benchmark on the BR municipal cycles. We note as a methodological observation that the architecture’s selective coverage at  $\tau = 0.40$  (§5.5) achieves 100% accuracy on the 7.9% most confident events, comparable in spirit to the high-confidence subset that drives Tetlock’s calibration claims.

*Social-media-based forecasting.* Tumasjan et al. (2010) launched a wave of Twitter-based vote-share predictions that subsequent reanalysis (Gayo-Avello 2012) showed could be made to predict any election by retroactive parameter selection. We exclude this paradigm from comparison: the leak-safe protocol used here would invalidate most of the published claims in that literature.

### 6.10 Differentiation from prior work

We summarize how the production architecture differs from five canonical references in the empirical-political-forecasting literature.

| Reference                                  | Mechanism                                                                                  | Output granularity              | Country / corpus                               |
|--------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------|
| Silver (2008-2024)<br>FiveThirtyEight      | weighted poll average + economic fundamentals                                              | state-day vote share            | US, 2008-2024                                  |
| Linzer (2013) JASA                         | dynamic Bayesian DLM (Stan, random-walk prior)                                             | state-day vote share            | US 1952-2008                                   |
| Hummel and Rothschild (2014) Elec. Studies | fundamentals only (no polls)                                                               | state vote share                | US                                             |
| Wang et al. (2015) IJF                     | full MRP over demographic strata, non-representative panel                                 | individual to state vote choice | US 2012 (Xbox)                                 |
| Heidemanns et al. (2020) HDSR              | DLM + explicit house-effects layer                                                         | state-day vote share            | US 2008-2020                                   |
| <b>This paper</b>                          | cohort empirical Bayes + closed-form Linzer + Laplace-smoothed (uf, regime) state baseline | event-level binary win          | BR + 11 cross-country, 12 cycles, 6,954 events |

Three structural differences. First, output granularity: prior work models continuous expected vote share at the state-day level; we predict the binary win indicator at the event-paired level (poll x candidate, with paired runner-up rows mirroring

the sign), which simplifies calibration to standard proper scoring rules and admits closed-form selective coverage. Second, prior parameterization: we condition the state-level prior on partisan regime (a five-class factor: left, pop\_left,

center, `pop_right`, `right`) rather than on demographic strata; this shrinks effective dimensionality from  $|U| \cdot |D|$  demographic cells to  $|U| \cdot |R|$  regime cells, allowing the prior to be Laplace-smoothed on the 394-event paired construction without microdata. Third, scope: the Brazilian core has no canonical aggregator-style precedent, and the eleven-country extension is, to our knowledge, the first leak-safe replication of an MRP-style state-level prior across electoral systems with binary, regime, and presidential / parliamentary heterogeneity. The architecture is intentionally more conservative than Stan-based hierarchical posteriors: closed-form Linzer plus a single  $w$  blend weight, fitted by leak-safe grid search, both fit and forecast in under three minutes on commodity hardware while remaining within 0.4 pp accuracy of the 0.9721 production point under repeated sensitivity analyses (§§6.5-6.8).

## 7 Limitations

Two concerns from earlier drafts (scalar Linzer drift, within-BR non-SP coverage) were tested empirically in §6.6 and §6.7 and resolved in favor of the production architecture; we therefore restrict this section to limitations the data does not refute. Two remain.

1. *Demographic poststratification.* The state baseline conditions on partisan regime, not on demographic strata. A full MRP in the sense of Gelman and Hill (2007) would weight by gender, education, income, and urban/rural strata using PNAD-Continua microdata; the relevant 4-trimestre 2025 release with post-2024-Census re-weighting was published by IBGE on 2026-03-27, so the data are operationally available. We defer the integration because the current 394-event paired construction does not carry per-event demographic strata, and ingesting them requires re-curating the seed CSVs and cross-country sources end to end.

2. *Eligibility filtering.* Jair Bolsonaro is filtered from 2026 forward predictions because of the Tribunal Superior Eleitoral ruling of 2023-06-30 (AIJE 0600814-85, 5-2, ineligible until 2030; Tribunal Superior Eleitoral 2023). The filter operates at the registry level rather than inside the model, so historical 2018 and 2022 events involving Bolsonaro remain in the training set and the model learns the partisan structure of pop-right candidates without conflating it with current 2026 viability. Should a higher court reverse the ruling before October 2026, the registry must be unfrozen and predictions recomputed; this contingent re-run rule is recorded in the supplementary pre-registration document.

## 8 Reproducibility

We provide a four-layer reproducibility stack: (i) a frozen code freeze with two git tags and a SHA-256 manifest covering source, input, and cached output files (supplementary), (ii) a deterministic single-seed pipeline that runs end-to-end in three minutes on commodity hardware, (iii) a containerized environment that lifts the host-Python dependency, and (iv) a smoke-test suite plus continuous integration on every push.

### 8.1 Code freeze, data provenance, and result hashes

The forecaster source code is published with the supplementary materials under two pre-registration git tags. The original freeze v1.2-prereg was created on 2026-05-07; a byte-equivalent re-freeze v1.3-prereg was created on 2026-05-08 with refreshed code SHA-256 values after non-substantive cleanup that does not alter the forecast snapshot, so both tags resolve to identical predictions on the frozen 2026 horizon. All polls are real, sourced from public Wikipedia poll-aggregation pages and FiveThirtyEight historical archives captured prior to the 2025-03-05 shutdown. Twenty source CSVs ship in the supplementary corpus, comprising the 394-event Brazilian core and the 6,954-event cross-country extension. Cached cross-validation JSONs ship alongside the source so byte-identical reproduction does not require re-running the grid. A SHA-256 manifest covering all source files, input CSVs, and cached result JSONs accompanies the supplementary archive; row-level entries are too granular for the body of the article and are omitted here.

### 8.2 Software environment

Python 3.11 with packages pinned in the supplementary requirements file:

```
numpy 1.26.x
scipy 1.11.x
pandas 2.1.x
scikit-learn 1.4.x
matplotlib 3.10.x
pymc-bart 0.11.x          (optional: BART
benchmark)
xgboost 3.2.x             (optional: ML
baseline)
fastapi 0.110.x
uvicorn 0.27.x
pydantic 2.x
weasyprint 60.x          (paper PDF
compilation)
```

A pinned `Dockerfile` at the repository root reproduces the environment in a container:



```
docker build -t vila-politica .
docker run --rm -v $(pwd)/data:/app/data vila-politica make reproduce
```

Random seed is 42 everywhere. Both `numpy.random` and the Python `random` module are seeded at script entry points.

### 8.3 One-command reproduction

A `Makefile` ships with the repository:

```
make smoke          # 29/29 contract tests, ~3
seconds
make stats          # bootstrap CIs + DM +
McNemar + Murphy
make baselines      # 5 ablation models year-fold
CV
make ml             # LogReg + RF + XGBoost + MLP
+ GaussianNB
make cross          # 12-cycle cross-country
make wsweep        # state-baseline weight sweep
+ figure
make hetero         # heteroscedastic Linzer
ablation experiment
make loso           # within-BR leave-one-state-
out experiment
make dropone        # drop-one-cycle robustness
experiment
make houseeffects   # per-institute bias
decomposition
make calibration    # reliability diagram + ECE/
MCE
make permutation    # permutation test on the MRP
gain
make autoresearch   # full 2,688-combo grid
search (verify v1.3 production)
make all            # everything above + paper
PDF rebuild
```

End-to-end reproduction takes approximately three minutes on a single modern x86 core (Intel i7-12700K, 16 GB RAM, no GPU). The dominant cost is the BART benchmark (~140 seconds for 6 folds at 200 trees, 150 draws); without BART, full reproduction is under one minute.

### 8.4 Smoke tests and continuous integration

The smoke-test suite runs all 29 contract tests covering the cohort fit, the state baseline, the blend formula, the year-fold CV harness, and the API endpoints. The repository GitHub Actions workflow runs the smoke test plus a full backtest reproduction on every push, on Python 3.11 and 3.12.

### 8.5 Pre-registration and blind protocol

The blend weight  $w$  and the baseline minimum-support threshold  $N \geq 3$  were specified before observing 2024 outcomes, with the formal pre-registration frozen on 2026-05-07. The pre-specified targets were 97% average and 85% on the

2024 SP fold; both were met (97.21% and 89.71%). Forward forecasts for the 2026 cycle, four locked hypotheses (H1 to H4), and the post-mortem evaluation protocol are recorded in `PREREGISTRATION.md`.

## 9 Conclusion

A Laplace-smoothed (UF, regime) baseline, computed only from out-of-fold training years and blended at  $w=0.36$  into a re-tuned cohort+Linzer ensemble, raises year-fold accuracy of the Vila INTEIA political forecaster from 94.16% (v1.2, no state baseline) to 97.21% (v1.3 production). The 2024 Sao Paulo mayoral fold, where every major polling firm had Boulos ahead and Nunes won by approximately three points, is recovered from 73.53% to 89.71% without leaking test outcomes and without overfitting any other cycle. Holding v1.3 hyperparameters fixed, the marginal contribution of the state baseline alone ( $w=0$  to  $w=0.36$ ) is +5.33 pp average and +25.00 pp on 2024 Sao Paulo.

The mechanism is a structurally simple proxy for MRP, applied over partisan-regime baselines per state rather than over demographic strata. Statistical significance is supported by a Diebold-Mariano test ( $DM=-4.92$ ,  $p=8.5e-7$ ) and a paired McNemar test ( $\chi^2=18.27$ ,  $p=1.9e-5$ ,  $b=22$ ,  $c=1$ ) on the 394-event year-fold series, plus a 2024 Sao Paulo per-fold test ( $DM=+7.79$ ,  $p=6.7e-15$ ; McNemar  $\chi^2=16.02$ ,  $p=6.3e-5$ ,  $b=17$ ,  $c=0$ ).

Cross-country replication on twelve cycles from eleven additional countries ( $n=6,954$  events: US 2016, 2020, 2022 midterms; UK 2019; FR 2022; AR 2023; BR 2014; DE 2021; MX 2024; TR 2023; IT 2022; IN 2024) demonstrates that the mechanism generalizes: in the eight single-uf cycles that exercise the state baseline directly, weighted accuracy moves from 97.86% to 100.00% and weighted Brier from 0.025 to 0.013. Exogenous state-level priors are thus a viable mechanism for absorbing cycle-specific industry-wide polling bias across electoral systems, ideological directions, and party landscapes. The architecture's transparency, its connection to ridge regression with state dummies, and its computable confidence intervals make it suitable for operational deployment alongside, rather than in place of, traditional aggregator-style models.

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## Author contributions

P.A.M. conceived the architecture, implemented the cohort empirical-Bayes and state-baseline modules, ran all benchmarks, and wrote the manuscript. I.M.V. implemented the curation pipeline for Brazilian and cross-country corpora, validated the leak-safe protocol, and reviewed every revision.

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This research received no external funding. All compute was performed on a single commodity workstation (Intel i7-12700K, 16 GB RAM, no GPU).

## Competing interests

The authors are co-founders of Vila INTEIA, which operates a public-facing forecasting product that uses an instance of the production v1.3 architecture described here. The product is non-commercial and the underlying source code is released under MIT license with the publication of this paper. The authors otherwise declare no competing interests.

## Data availability statement

All data and code are public. The 394-event Brazilian core, the 6,954-event cross-country extension, the cached cross-validation result JSONs, and the pre-registration manifest are released under MIT license in the supplementary archive. A frozen Docker container reproduces every figure and table reported in this paper end to end in approximately three minutes on commodity hardware. Two pre-registration git tags (`v1.2-prereg`, `v1.3-prereg`) anchor the empirical claims to byte-identical artefacts.

## Replication materials

Source code, polling CSVs, qualitative pool, intermediate JSONs, the pre-registration document, and a SHA-256 manifest are archived alongside the published article. The Makefile's `make all` target reproduces every numerical claim in the manuscript from raw inputs.

## Ethics statement

This research relies exclusively on publicly available aggregate-level polling data, certified electoral outcomes, and publicly published prediction-market prices. No individual-level data and no IRB-eligible human subjects research was involved.

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## Appendix A. Hyperparameter grid

Full search grid for the baseline ensemble and the state-baseline blend weight.

Baseline ensemble (2,688 points):

- $\text{stein\_shrink}$ : {0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40} (8 values)
- $w_{\text{linzer}}$ : {0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 1.00} (8 values)
- $\sigma_0$  (pp): {3.0, 4.0, 5.0, 6.0, 7.0, 8.0} (6 values)
- $\sigma_1$  (pp/day): {0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08} (7 values)

State-baseline blend (15 points):  $w$  in {0.00, 0.10, 0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.32, 0.35, 0.36, 0.37, 0.38, 0.40}.

Auxiliary thresholds (fixed, not searched):

- $\text{min\_n}$  (per (uf, regime) cell): 3
- Laplace  $\alpha = \beta = 1$
- Operational filter  $T \leq 30$  days

Selection criterion: maximize average year-fold accuracy weighted by per-cycle event count, with a tiebreak on minimum 2024 SP accuracy. The chosen production v1.3 point ( $s=0.40$ ,  $w_{\text{linzer}}=0.70$ ,  $\sigma_0=3.0$ ,  $\sigma_1=0.01$ ,  $w=0.36$ ) is unique under this criterion across the joint  $2,688 \times 15 = 40,320$  candidate configurations. The pre-MRP v1.2 best ( $s=0.05$ ,  $w_{\text{linzer}}=0.50$ ,  $\sigma_0=4.0$ ,  $\sigma_1=0.05$ ,  $w=0$ ) is the unique optimum under the same criterion when  $w$  is constrained to 0.

## Appendix B. Implicit hierarchical model

The blend  $p_{\text{final}} = (1 - w) [(1 - w_t) \tilde{p}_k + w_t \Phi(\ell/\sigma(d))] + w p_{u,r}$  is a point-estimator analogue of a unified hierarchical Bayesian model. In that model, the cohort base rate  $p_k$  would receive a Stein-shrunk Beta prior centred on the global rate; the state baseline  $p_{u,r}$  would receive an informationless  $\text{Beta}(1, 1)$  prior consistent with our Laplace smoothing with min-N threshold three; the Linzer drift parameters  $\sigma_0, \sigma_1$  would receive truncated-normal hyperpriors anchored at the values selected by autoresearch grid search; and the binary outcome  $y$  would follow  $\text{Bernoulli}(p_{\text{final}})$ . Our production estimator replaces full posterior inference (Hamiltonian Monte Carlo over this structure) with point estimates: the cohort rate is the Stein-shrunk maximum-likelihood estimate, the state baseline is the Laplace-smoothed contingency mean, and  $(\sigma_0, \sigma_1, s, w_t, w)$  are selected by grid search. The fully Bayesian implementation is left for follow-up work; the point estimator captures the same conditional structure with substantially lower compute cost and matches the  $\text{Beta}(1, 1)$  Laplace prior on the state baseline exactly.

## Appendix C. Pre-registration timestamp

| Tag                       | Frozen on  | HEAD                                                 | Forecast snapshot                                                       | Purpose                                                                                                                                                                                                 |
|---------------------------|------------|------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>v1.2-pre-reg</code> | 2026-05-07 | 7d2403b7dd575<br>6f95b378331e4<br>3405cae60e62d<br>a | the frozen 2026 forecast<br>snapshot (supplementary)<br>SHA 9e693389... | Original freeze concurrent with the OSF/arXiv pre-registration.                                                                                                                                         |
| <code>v1.3-pre-reg</code> | 2026-05-08 | 4fc1456ca20db<br>2bd028939c735<br>80159d91018ba<br>9 | identical to <code>v1.2-prereg</code>                                   | Refrozen with refreshed code-file SHAs after editorial cleanup. Forecast snapshot is byte-identical to <code>v1.2-prereg</code> ; only the engine and predict-script SHAs in the SHA-256 table changed. |

The SHA-256 tables for both tags are recorded in the supplementary pre-registration document §3. The shell-command procedure for creating `v1.3-prereg` is in the pre-registration freeze procedure (supplementary) §4b. The original `v1.2-prereg` tag remains in git history; nothing was rewritten.

$WBV| < 8e-3$  across all six cycles under quantile bins (perfect identity on 2010, 2018, 2020, 2022 federal; small bin-edge residual on 2016 SP and 2024 SP, the standard finite-k binning artifact discussed in Brier (2009), QJRM 135, 1512-1519). Per-cycle and pooled WBV components are stored in the statistical-rigor result file (supplementary) under the `murphy_pooled` and `murphy_per_cycle` keys. [↩](#)

1. For finite  $k$  the textbook identity  $BS = REL - RES + UNC$  holds only up to a within-bin variance term  $WBV = E[Var(p | bin)]$ . We report REL, RES, UNC as defined in Murphy (1973) and observe  $|BS - (REL - RES + UNC +$